



J.J. College of Engineering and Technology

Department of Mechanical Engineering

ME25CO8

Metallurgy and Materials Science

Student Notes

Unit - I: Constitution of Alloys and Phase Diagrams

Scheduled Hours: 9 Periods

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Course Handling Faculty

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Contents

Course Objectives	iii
Full Course Content	iii
Course Outcomes	iv
Unit I Learning Focus	v
Engineering Context, Indian Relevance and Innovation Link	v
Text Books	vi
References	vii
1 Introduction to Unit I	1
2 Constitution of Alloys	1
2.1 Alloy	1
2.2 Constitution of an Alloy	2
2.3 Need for Alloying	2
2.4 Basic Terms	2
3 Solid Solutions	2
3.1 Types of Solid Solutions	3
3.2 Substitutional Solid Solution	3
3.3 Interstitial Solid Solution	4
3.4 Substitutional and Interstitial Solid Solutions - Comparison	4
4 Phase Diagrams	5
4.1 Definition	5
4.2 Importance of Phase Diagrams	5
4.3 Important Terms Used in Phase Diagrams	5
4.4 Gibbs Phase Rule	6
4.5 Lever Rule	6
5 Isomorphous Phase Diagram	8
5.1 Definition	8
5.2 Features of Isomorphous System	8
5.3 Cooling of an Isomorphous Alloy	8

6	Invariant Reactions in Phase Diagrams	9
6.1	Eutectic Reaction	9
6.2	Eutectoid Reaction	9
6.3	Peritectic Reaction	10
6.4	Peritectoid Reaction	11
6.5	Comparison of Invariant Reactions	11
7	Iron–Iron Carbide Equilibrium Diagram	12
7.1	Importance	12
7.2	Important Composition Ranges	12
7.3	Important Temperatures and Points	12
7.4	Phases in the Iron–Carbon System	13
7.5	Microstructures of Plain Carbon Steels	13
8	Classification of Steels	14
8.1	Definition of Steel	14
8.2	Classification Based on Carbon Content	15
8.3	Microstructure, Properties and Applications of Steels	15
9	Classification of Cast Irons	16
9.1	Definition of Cast Iron	16
9.2	Gray Cast Iron	17
9.3	White Cast Iron	18
9.4	Chilled Cast Iron	19
9.5	Ductile Cast Iron	19
9.6	Malleable Cast Iron	20
9.7	Cast Irons - Comparison	20
10	Important Exam-Oriented Points	21
10.1	Very Short Answer Points	21
10.2	Important Two-Mark Questions	22
10.3	Important Five-Mark Questions	22
10.4	Important Long-Answer Questions	22
11	Unit Summary	23
	Glossary of Important Terminologies	24

Course Objectives

1. To learn the construction of phase diagrams and the use of the Iron–Iron carbide phase diagram for understanding microstructure formation.
2. To learn the selection and application of various heat treatment processes and their influence on microstructure formation.
3. To illustrate the different types of ferrous and non-ferrous alloys and their uses in the engineering field.
4. To illustrate different polymers, ceramics and composites and their uses in the engineering field.
5. To learn various testing procedures and failure mechanisms in engineering materials.

Full Course Content

Course Code: ME25CO8

Course Title: Metallurgy and Materials Science

Unit	Content
UNIT I	CONSTITUTION OF ALLOYS AND PHASE DIAGRAMS Constitution of alloys – Solid solutions, substitutional and interstitial phase diagrams, isomorphous, eutectic, eutectoid, peritectic, and peritectoid reactions, Iron–Iron carbide equilibrium diagram. Classification of steel and cast-iron microstructure, properties and application.
UNIT II	HEAT TREATMENT Definition – Full annealing, stress relief, recrystallisation and spheroidising – normalizing, hardening and tempering of steel. Isothermal transformation diagrams – cooling curves superimposed on I.T. diagram – continuous cooling Transformation (CCT) diagram – Austempering, Martempering – Hardenability, Jominy end quench test – case hardening, carburizing, nitriding, cyaniding, carbonitriding – Flame and Induction hardening – Vacuum and Plasma hardening – Thermo-mechanical treatments – elementary ideas on sintering.
UNIT III	FERROUS AND NON-FERROUS METALS

Effect of alloying additions on steel (Mn, Si, Cr, Mo, Ni, V, Ti and W) – stainless and tool steels – HSLA – Maraging steels – grey, white, malleable, spheroidal alloy cast irons, copper and its alloys – brass, bronze and cupronickel – aluminium and its alloys; Al–Cu precipitation strengthening treatment – titanium alloys, Mg-alloys, Ni-based super alloys – shape memory alloys – properties and applications – overview of materials standards.

UNIT IV NON-METALLIC MATERIALS

Polymers – types of polymers, commodity and engineering polymers – properties and applications of PE, PP, PS, PVC, PMMA, PET, PC, PA, ABS, PAI, PPO, PPS, PEEK, PTFE, thermoset polymers, urea and phenol formaldehydes – nylon, engineering ceramics – properties and applications of Al_2O_3 , SiC, Si_3N_4 , PSZ and SIALON – intermetallics – composites – matrix and reinforcement materials – applications of composites – nano composites.

UNIT V MECHANICAL PROPERTIES AND DEFORMATION MECHANISMS

Mechanisms of plastic deformation, slip and twinning – types of fracture – fracture mechanics – Griffith's theory – testing of materials under tension, compression and shear loads – hardness tests (Brinell, Vickers and Rockwell), micro and nano-hardness tests, impact test Izod and Charpy, fatigue and creep failure mechanisms.

Total: 45 periods (9 periods each)

Course Outcomes

At the end of the course, the students would be able to:

1. Explain alloys and phase diagrams, the Iron–Iron carbon diagram and steel classification.
2. Explain isothermal transformation, continuous cooling diagrams and different heat treatment processes.
3. Clarify the effect of alloying elements on ferrous and non-ferrous metals.
4. Summarize the properties and applications of non-metallic materials.
5. Explain the testing of mechanical properties.

Unit I Learning Focus

For examination preparation, Unit I should be studied with the following sequence:

1. Basic terms: alloy, phase, component, system, equilibrium and microstructure.
2. Solid solutions: substitutional and interstitial solid solutions with examples.
3. Phase diagrams: liquidus, solidus, solvus, tie line, lever rule and invariant reactions.
4. Iron–Iron carbide diagram: important phases, reactions, temperatures and carbon percentages.
5. Classification of steels and cast irons with microstructure, properties and applications.

Engineering Context, Indian Relevance and Innovation Link

Why this unit matters: Phase diagrams and alloy constitution are not only examination topics. They are the basic tools used to decide composition, processing temperature, microstructure and service properties of engineering alloys.

Indian iron and steel usage: classroom facts

- India is among the leading steel producing nations, and recent industry reporting identifies India as the world's second-largest crude steel producer.
- The national steel capacity target is commonly discussed around 300 million tonnes by 2030–31; therefore, metallurgy knowledge is directly connected with India's manufacturing, construction and infrastructure growth.
- Iron and steel are essential for bridges, buildings, railway tracks, automobiles, agricultural equipment, pressure vessels, power plants, machine tools and defence structures.
- The quality of steel used in infrastructure depends on microstructure control. For example, ferrite-pearlite steels are used in many structural applications, pearlitic steels are important in rails and wires, and alloy steels are used where higher strength or wear resistance is required.
- Cast irons remain important in machine beds, engine blocks, brake components, pump housings and heavy machinery because of good castability, vibration damping and machinability.

Start-up and project ideas linked with Unit I

1. **Microstructure identification support:** A simple mobile or web tool that helps students and small workshops identify ferrite, pearlite, cementite, graphite flakes and graphite nodules from microscope images.
2. **Steel selection assistant:** A database-based application that suggests low-carbon steel, medium-carbon steel, high-carbon steel or cast iron based on load, wear, weldability and cost requirements.
3. **Foundry quality checklist:** A digital checklist for small foundries to connect casting defects, cooling rate, carbon equivalent and expected cast iron microstructure.
4. **Phase diagram learning kit:** A classroom kit or interactive simulator to teach lever rule, tie line and invariant reactions using simple alloy systems.
5. **Infrastructure material awareness:** A low-cost training module for diploma/engineering students explaining how steel grade, microstructure and heat treatment affect buildings, bridges and railways.

High-technology facts related to Unit I

- Advanced steels used in automobiles are designed by controlling phases such as ferrite, martensite, bainite and retained austenite.
- Electrical steels used in motors, transformers and electric vehicles require careful control of composition, grain structure and magnetic properties.
- Turbine, aerospace and energy components use alloy design principles similar to phase diagram interpretation, especially for controlling precipitates and high-temperature phases.
- Additive manufacturing of metals also depends on rapid solidification, phase transformation and microstructure control.
- Materials informatics and AI-based alloy design use databases of composition, phases and properties; therefore, the basic ideas of this unit are useful even in modern computational metallurgy.

Text Books

1. Kenneth G. Budinski and Michael K. Budinski, *Engineering Materials*, Prentice Hall of India Private Limited, 9th Edition, 2018.

2. Sydney H. Avner, *Introduction to Physical Metallurgy*, McGraw Hill Book Company, 1994.

References

1. A. Alavudeen, N. Venkateshwaran, and J. T. Winowlin Jappes, *A Textbook of Engineering Materials and Metallurgy*, Laxmi Publications, 2006.
2. Amandeep Singh Wadhwa and Harvinder Singh Dhaliwal, *A Textbook of Engineering Material and Metallurgy*, University Sciences Press, 2008.
3. G. S. Upadhyay and Anish Upadhyay, *Materials Science and Engineering*, Viva Books Pvt. Ltd., New Delhi, 2020.
4. V. Raghavan, *Materials Science and Engineering*, Prentice Hall of India Pvt. Ltd., 6th Edition, 2019.
5. William D. Callister, *Materials Science and Engineering*, Wiley India Pvt. Ltd., 2nd Edition Reprint, 2019.

1 Introduction to Unit I

Materials used in engineering applications are rarely used in the perfectly pure form. Metals are commonly mixed with other elements to improve strength, hardness, corrosion resistance, castability, machinability, wear resistance and service life. The study of alloys and phase diagrams helps engineers understand how composition and temperature affect microstructure and properties.

Exam point: In examination answers, always connect **composition** → **microstructure** → **properties** → **applications**.

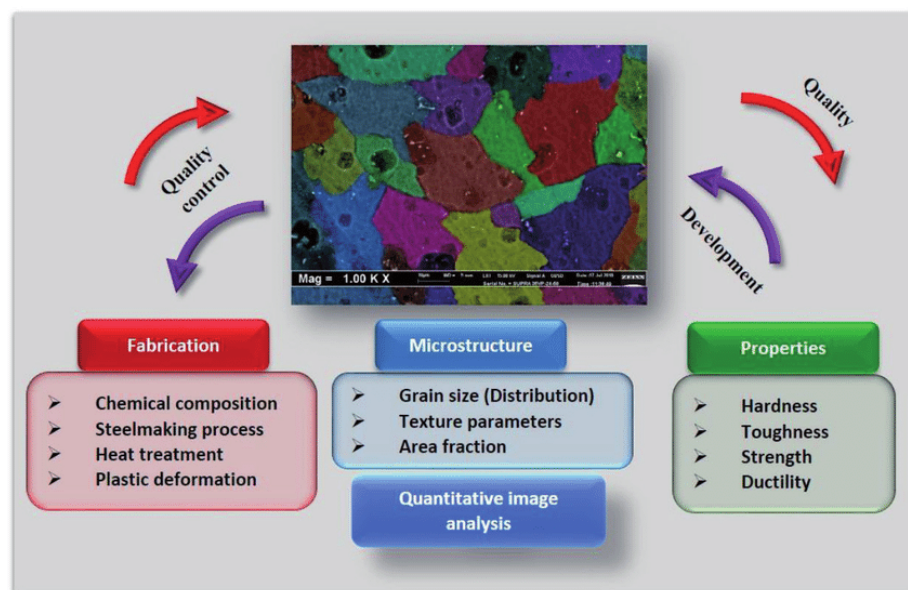


Figure 1: Relationship between alloy composition, microstructure, properties and applications.

2 Constitution of Alloys

2.1 Alloy

An **alloy** is a metallic material formed by combining two or more elements, of which at least one is a metal. The added element is called an alloying element. Alloying is done to improve the properties of the base metal.

Examples:

- Steel: Iron + Carbon
- Brass: Copper + Zinc
- Bronze: Copper + Tin
- Duralumin: Aluminium + Copper + Magnesium + Manganese

2.2 Constitution of an Alloy

The **constitution of an alloy** refers to the phases, microstructures and chemical distribution present in the alloy under a given condition of temperature, pressure and composition.

2.3 Need for Alloying

Alloying is carried out to:

1. Increase strength and hardness.
2. Improve corrosion and oxidation resistance.
3. Improve wear resistance.
4. Improve castability and machinability.
5. Improve high-temperature properties.
6. Modify microstructure and phase transformation behaviour.

2.4 Basic Terms

Term	Meaning
Solvent	The major element present in an alloy.
Solute	The minor element dissolved in the solvent.
Solid solution	A single solid phase formed when solute atoms dissolve in the solvent lattice.
Phase	A physically homogeneous and mechanically separable part of a system.
Microstructure	Arrangement of phases observed under a microscope.
Equilibrium	Stable condition in which no further change occurs with time at fixed temperature and pressure.

3 Solid Solutions

A **solid solution** is a solid phase in which atoms of one element are dissolved in the crystal lattice of another element. The base metal is the solvent and the dissolved element is the solute.

3.1 Types of Solid Solutions

Solid solutions are mainly classified into:

1. Substitutional solid solution
2. Interstitial solid solution

3.2 Substitutional Solid Solution

A **substitutional solid solution** is formed when solute atoms replace solvent atoms in the crystal lattice.

Conditions favouring substitutional solid solution:

1. Atomic size difference should usually be less than about 15%.
2. Solute and solvent should have the same crystal structure.
3. Electronegativity difference should be small.
4. Valency should be similar.

Example: Copper–Nickel (Cu–Ni). Both Cu and Ni have FCC crystal structure and similar atomic radii, hence they form a complete solid solution.

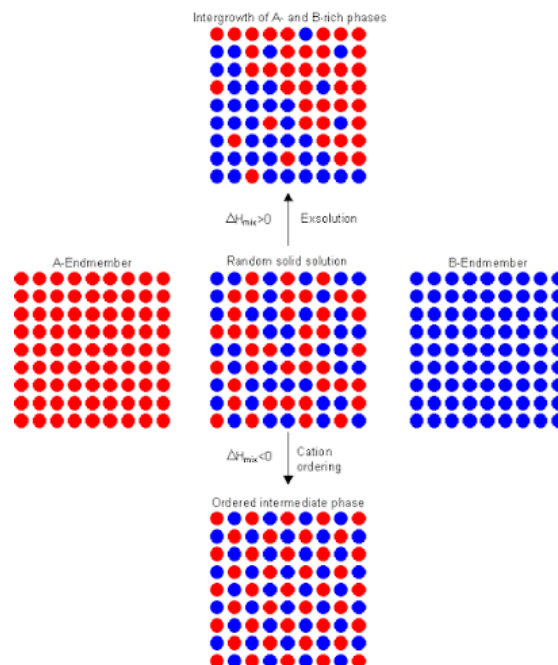


Figure 2: Substitutional solid solution showing replacement of solvent atoms by solute atoms.

3.3 Interstitial Solid Solution

An **interstitial solid solution** is formed when small solute atoms occupy the interstitial spaces or voids between larger solvent atoms.

Examples:

- Carbon in iron
- Nitrogen in iron
- Hydrogen in metals

In ferrous alloys, carbon forms interstitial solid solutions with iron. Ferrite and austenite are important interstitial solid solutions in the Iron–Carbon system.

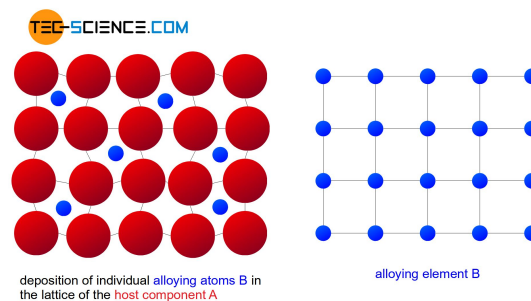


Figure 3: Interstitial solid solution showing small atoms occupying the spaces between larger atoms.

3.4 Substitutional and Interstitial Solid Solutions - Comparison

Point	Substitutional	Interstitial
Atomic position	Solute atoms replace solvent atoms	Solute atoms occupy interstitial spaces
Solute size	Similar to solvent atoms	Much smaller than solvent atoms
Examples	Cu–Ni, Cu–Zn	C in Fe, N in Fe
Effect	Moderate lattice distortion	High lattice distortion
Importance	Common in many alloy systems	Very important in steels

Exam point: For a short-answer question, write the definition, draw a simple lattice sketch and give one example for each type.

4 Phase Diagrams

4.1 Definition

A **phase diagram** is a graphical representation that shows the phases present in a material system at different temperatures, compositions and pressures. For most engineering alloys, pressure is constant; therefore, binary phase diagrams are usually plotted between **temperature** and **composition**.

4.2 Importance of Phase Diagrams

Phase diagrams are useful to:

1. Identify phases present at a given temperature and composition.
2. Determine phase transformation temperatures.
3. Predict microstructure after solidification or cooling.
4. Calculate the amount of each phase using the lever rule.
5. Select suitable alloy composition for engineering applications.

4.3 Important Terms Used in Phase Diagrams

Term	Meaning
Component	Independent chemical constituent of a system.
Phase	Homogeneous part of a system with uniform composition and properties.
Liquidus line	Line above which the alloy is completely liquid.
Solidus line	Line below which the alloy is completely solid.
Solvus line	Line showing limit of solid solubility.
Tie line	Horizontal line drawn in a two-phase region to find phase compositions.
Invariant reaction	Reaction occurring at fixed temperature and composition with three phases in equilibrium.

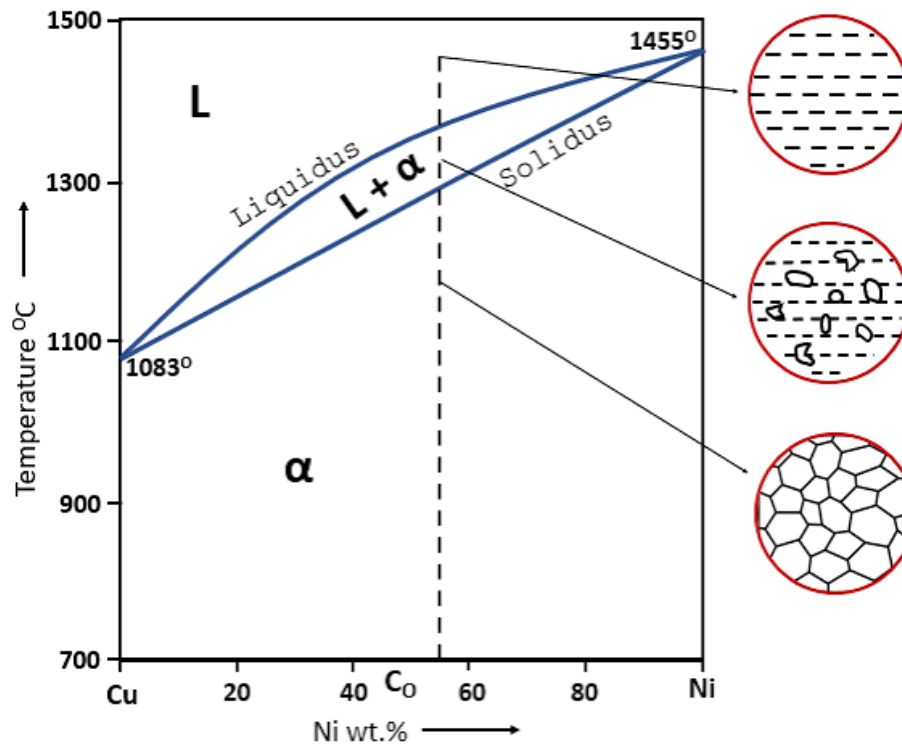


Figure 4: Basic binary phase diagram showing liquidus, solidus, single-phase and two-phase regions.

4.4 Gibbs Phase Rule

Gibbs phase rule gives the relationship between number of phases, components and degrees of freedom:

$$P + F = C + 2 \quad (1)$$

where

- P = number of phases
- F = degrees of freedom
- C = number of components

For condensed systems at constant pressure, the phase rule becomes:

$$P + F = C + 1 \quad (2)$$

4.5 Lever Rule

In a two-phase region, the amount of each phase can be found using the **lever rule**. A horizontal tie line is drawn at the required temperature. The compositions at the two ends of the tie line are used for calculation.

For a two-phase region containing solid (S) and liquid (L):

$$W_S = \frac{C_L - C_0}{C_L - C_S} \quad (3)$$

$$W_L = \frac{C_0 - C_S}{C_L - C_S} \quad (4)$$

where C_0 is the alloy composition, C_S is the composition of solid and C_L is the composition of liquid.

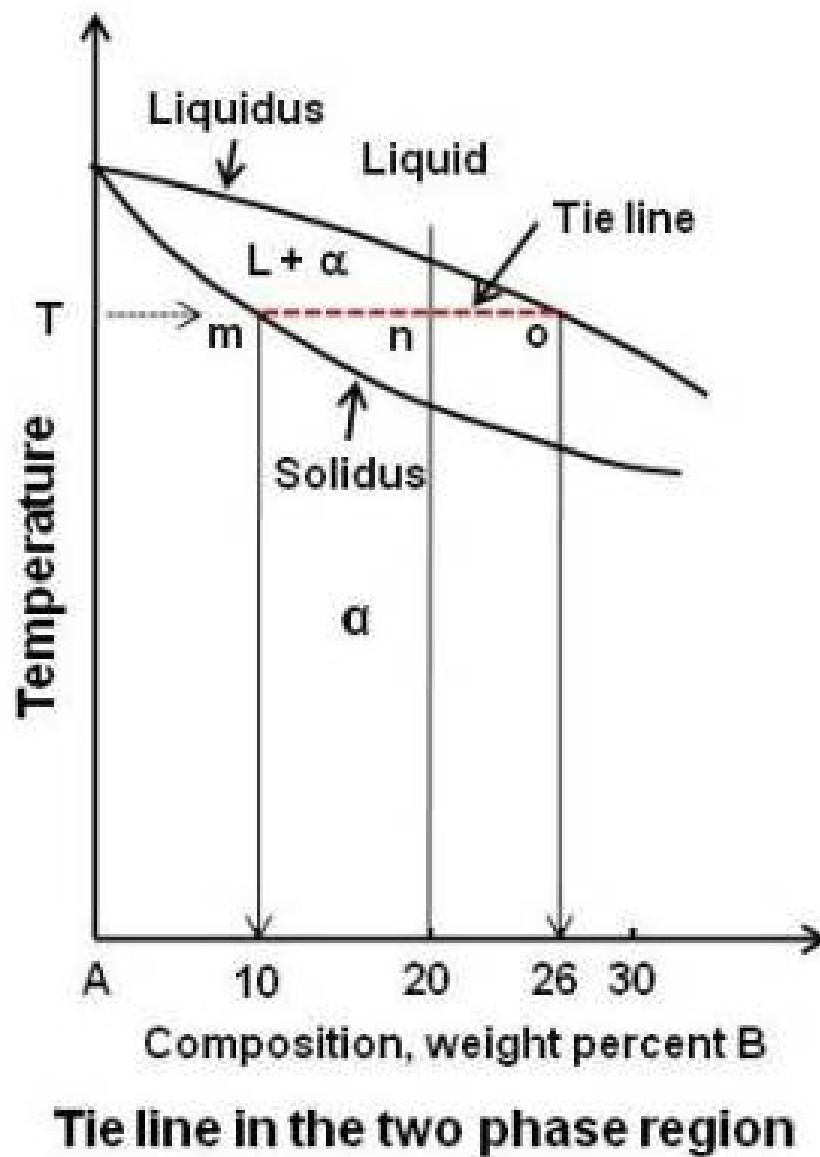


Figure 5: Tie line and lever rule in a binary phase diagram.

Exam point: In lever-rule problems, first draw the tie line, mark C_S , C_L and C_0 , then apply the opposite-arm principle.

5 Isomorphous Phase Diagram

5.1 Definition

An **isomorphous system** is a binary alloy system in which the two components are completely soluble in each other in both liquid and solid states.

Example: Cu–Ni system.

5.2 Features of Isomorphous System

1. Complete liquid solubility.
2. Complete solid solubility.
3. One solid phase α is formed over the entire composition range.
4. Liquidus and solidus lines separate liquid, liquid + solid and solid regions.

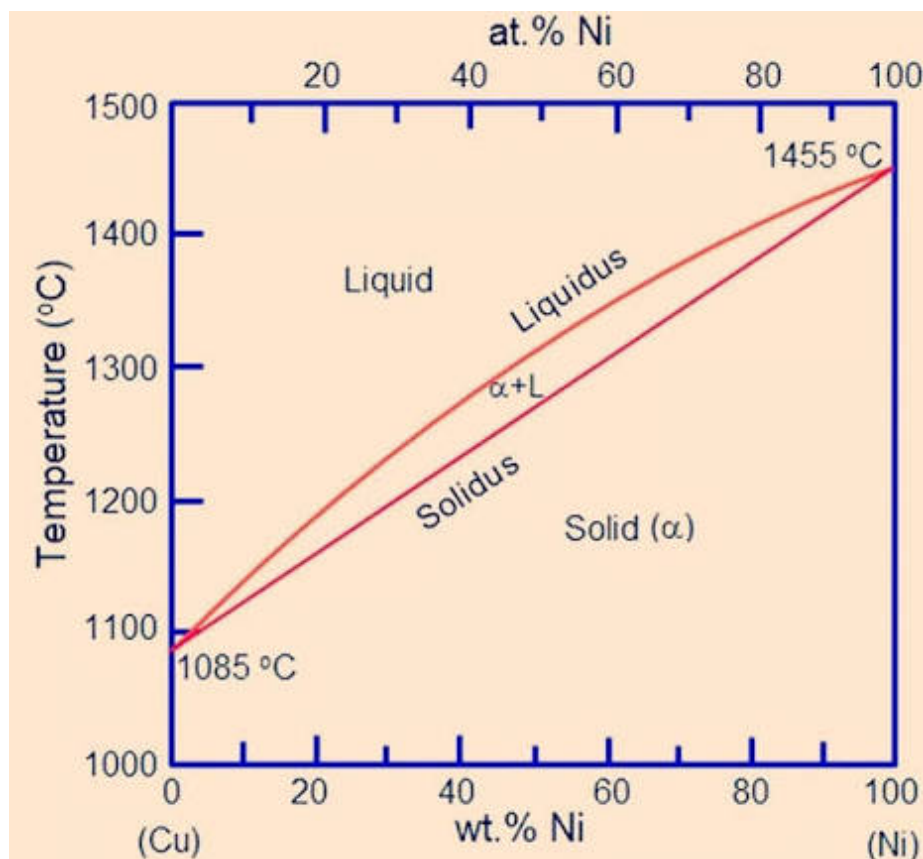


Figure 6: Cu–Ni isomorphous phase diagram.

5.3 Cooling of an Isomorphous Alloy

When an isomorphous alloy is cooled from the liquid state:

1. Above liquidus: alloy is completely liquid.
2. Between liquidus and solidus: liquid + solid phases coexist.
3. Below solidus: alloy becomes completely solid solution.

6 Invariant Reactions in Phase Diagrams

Invariant reactions are reactions that occur at a fixed temperature and composition. In binary alloy systems, important invariant reactions include eutectic, eutectoid, peritectic and peritectoid reactions.

6.1 Eutectic Reaction

A **eutectic reaction** is a reaction in which one liquid phase transforms into two solid phases during cooling.



General example: Pb–Sn solder alloy.

Iron–Carbon example: Liquid transforms to austenite and cementite at about 4.3 wt.% C and 1147°C.

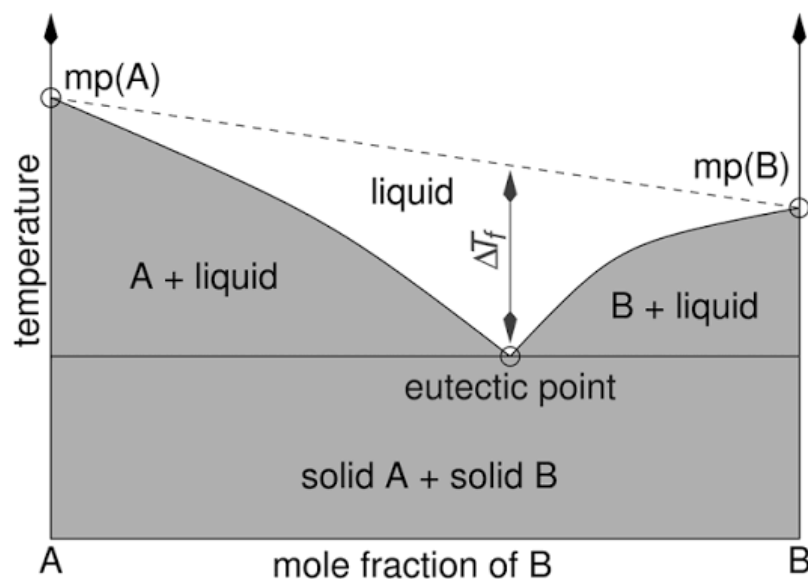


Figure 7: Schematic representation of eutectic reaction.

6.2 Eutectoid Reaction

A **eutectoid reaction** is a reaction in which one solid phase transforms into two different solid phases during cooling.



In the Iron–Carbon system, austenite transforms into ferrite and cementite at about 0.8 wt.% C and 727°C. The resulting lamellar mixture is called **pearlite**.

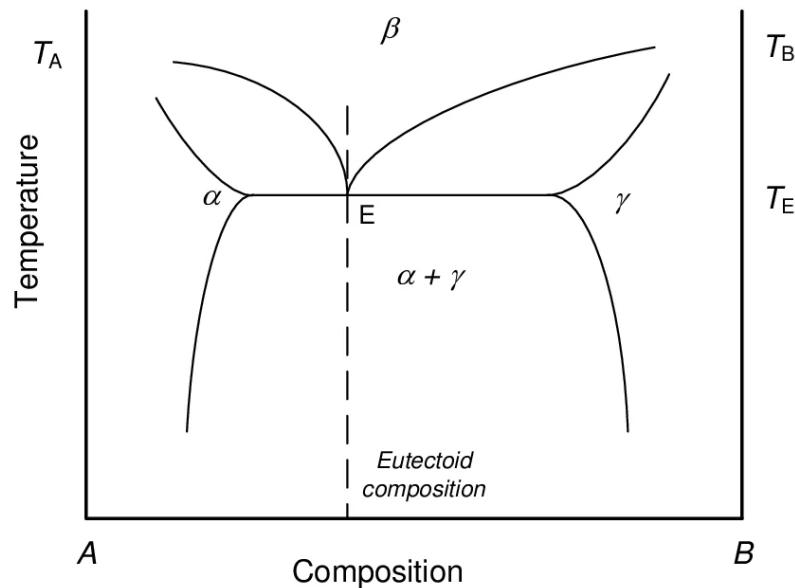


Figure 8: Schematic representation of eutectoid reaction.

6.3 Peritectic Reaction

A **peritectic reaction** is a reaction in which a liquid phase and a solid phase react to form a new solid phase during cooling.



In the Iron–Carbon system, liquid and delta ferrite react to form austenite near 1493°C.

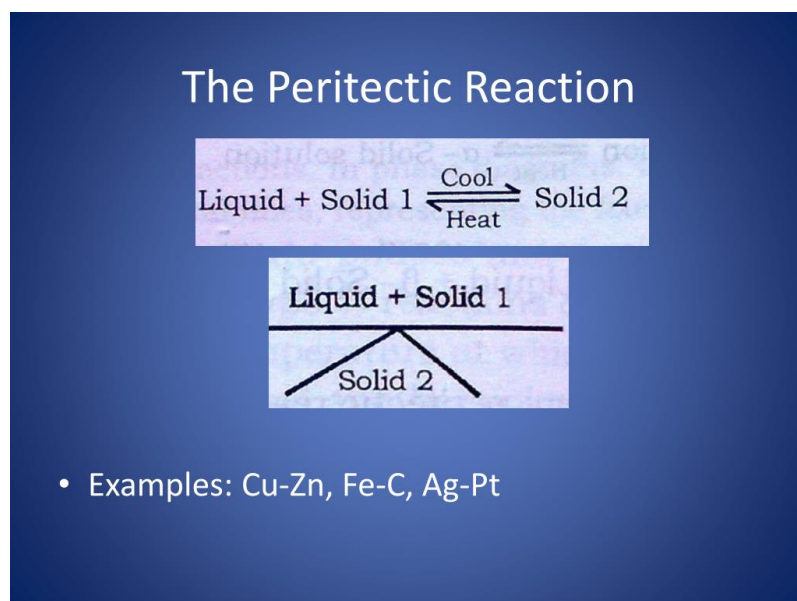


Figure 9: Schematic representation of peritectic reaction.

6.4 Peritectoid Reaction

A **peritectoid reaction** is a reaction in which two solid phases react to form a new solid phase during cooling.

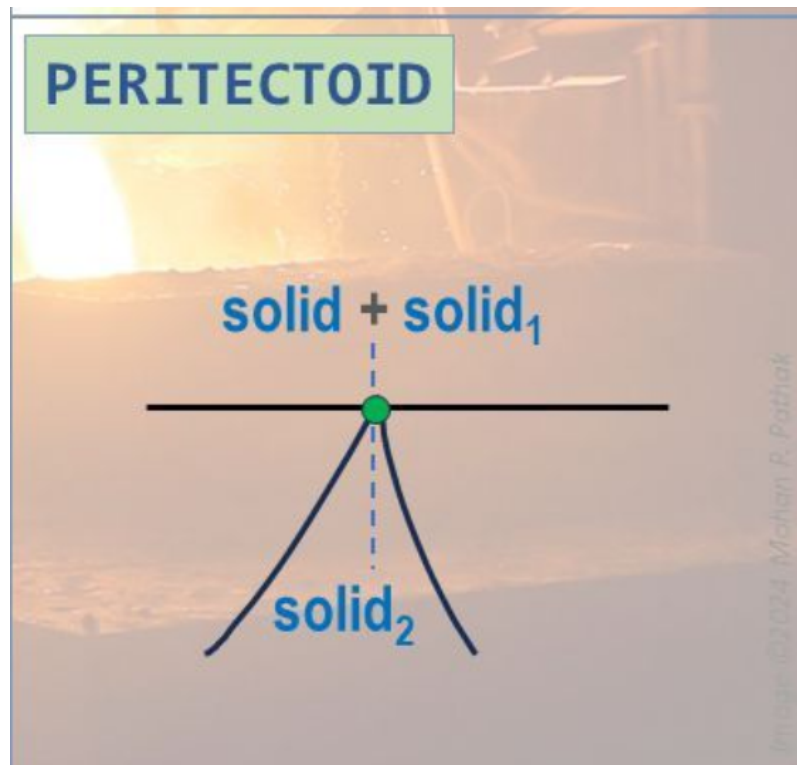


Figure 10: Schematic representation of peritectoid reaction.

6.5 Comparison of Invariant Reactions

Reaction	General equation on cooling	Main idea	Example
Eutectic	$L \rightarrow \alpha + \beta$	Liquid becomes two solids	Pb-Sn, Fe-C
Eutectoid	$\gamma \rightarrow \alpha + \beta$	One solid becomes two solids	Pearlite formation
Peritectic	$L + \alpha \rightarrow \beta$	Liquid + solid become new solid	Fe-C
Peritectoid	$\alpha + \beta \rightarrow \gamma$	Two solids become new solid	Alloy systems

Exam point: For invariant reactions, equations are very important. Memorise the phase-change direction during cooling.

7 Iron–Iron Carbide Equilibrium Diagram

7.1 Importance

The Iron–Iron carbide diagram is the most important phase diagram for ferrous materials. It is used to understand the microstructure and properties of steels and cast irons.

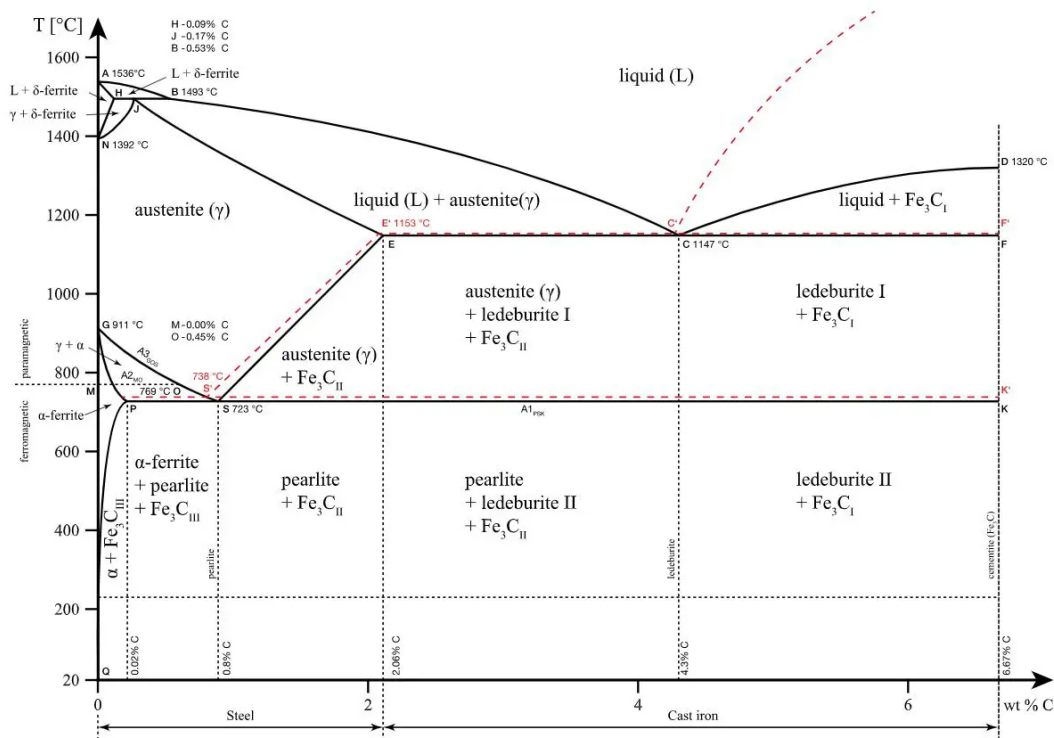


Figure 11: Iron–Iron carbide equilibrium diagram.

7.2 Important Composition Ranges

Composition range	Material classification
0 to about 0.02 wt.% C	Very low carbon solid solution / ferrite region
0.02 to 2.11 wt.% C	Steels
2.11 to 6.67 wt.% C	Cast irons
6.67 wt.% C	Cementite composition, Fe ₃ C

7.3 Important Temperatures and Points

Point/Temperature	Significance
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A_1 / eutectoid temperature	About 727°C; austenite transforms to pearlite.
A_3 line	Boundary between ferrite + austenite and austenite regions for hypo-eutectoid steels.
A_{cm} line	Boundary between austenite and austenite + cementite regions for hyper-eutectoid steels.
Eutectoid composition	About 0.8 wt.% C.
Eutectic composition	About 4.3 wt.% C.
Maximum carbon in austenite	About 2.11 wt.% C at 1147°C.
Cementite composition	6.67 wt.% C.

7.4 Phases in the Iron–Carbon System

Phase	Description	Property
Ferrite (α)	Interstitial solid solution of carbon in BCC iron	Soft, ductile, magnetic below Curie temperature
Austenite (γ)	Interstitial solid solution of carbon in FCC iron	Non-magnetic, high carbon solubility, stable at high temperature
Cementite (Fe_3C)	Iron carbide containing 6.67 wt.% C	Very hard and brittle
Pearlite	Lamellar mixture of ferrite and cementite	Medium strength and hardness
Ledeburite	Eutectic mixture in cast iron region	Hard and brittle constituent

7.5 Microstructures of Plain Carbon Steels

Steel type	Carbon content	Room-temperature microstructure under slow cooling
Hypo-eutectoid steel	Less than 0.8 wt.% C	Proeutectoid ferrite + pearlite
Eutectoid steel	About 0.8 wt.% C	Pearlite

Hyper-eutectoid steel	0.8 to 2.11 wt.% C	Proeutectoid cementite + pearlite
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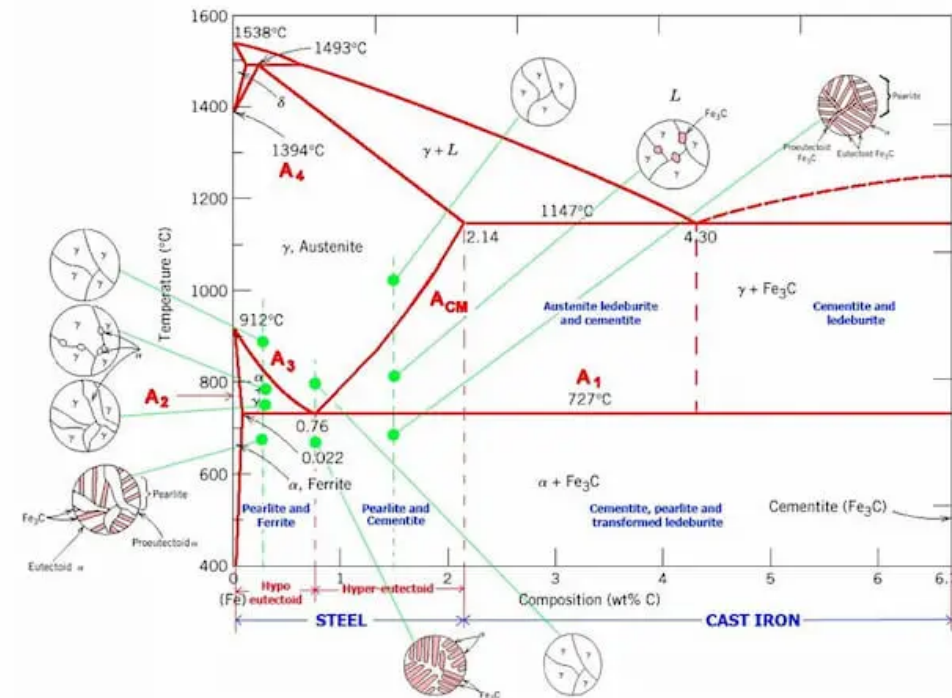


Figure 12: Microstructures of hypo-eutectoid, eutectoid and hyper-eutectoid steels.

Exam point: A very common exam question is: “Explain the Iron–Iron carbide equilibrium diagram with phases and invariant reactions.” Draw the diagram, mark 0.8% C, 2.11% C, 4.3% C, 6.67% C, 727°C and 1147°C.

8 Classification of Steels

8.1 Definition of Steel

Steel is an alloy of iron and carbon containing up to about 2.11 wt.% carbon. It may also contain elements such as manganese, silicon, nickel, chromium, molybdenum and vanadium to improve specific properties.

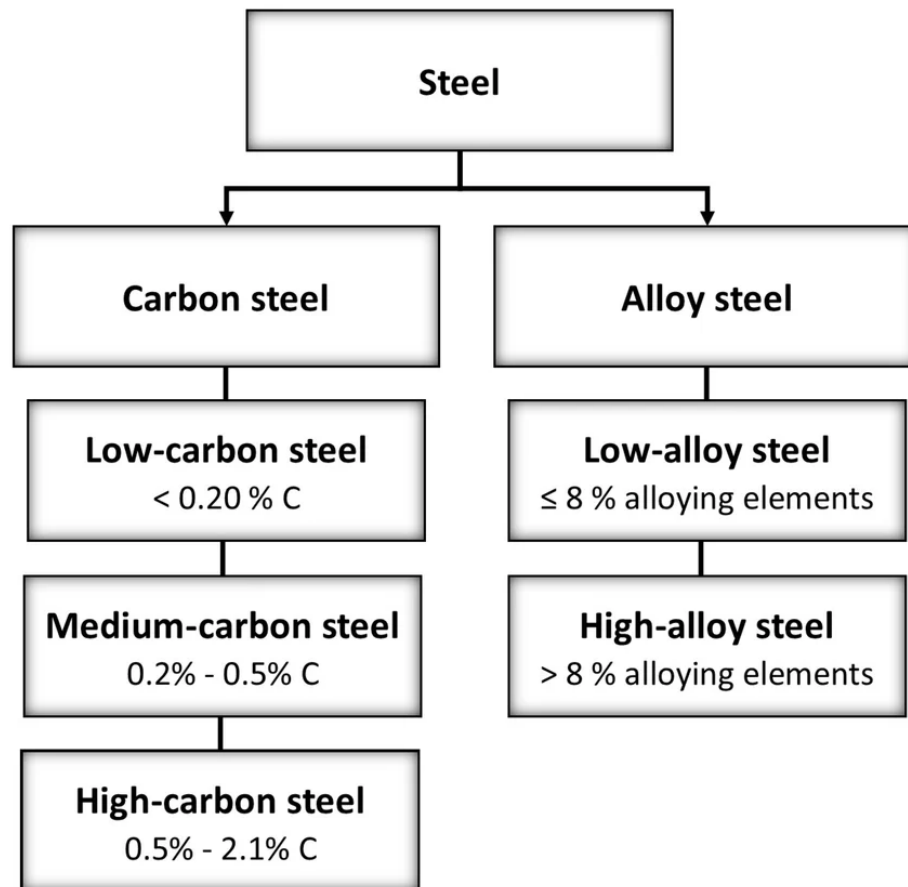


Figure 13: Classification of steels based on carbon content and alloying elements.

8.2 Classification Based on Carbon Content

Type	Carbon content	General properties	Applications
Low-carbon steel	0.05–0.30% C	Soft, ductile, weldable, easily formable	Sheets, automobile panels, wires, structural sections
Medium-carbon steel	0.30–0.60% C	Higher strength and hardness, moderate ductility	Shafts, gears, axles, crankshafts, rails
High-carbon steel	0.60–1.00% C	High hardness, high wear resistance, lower ductility	Springs, cutting tools, high-strength wires
Ultra-high-carbon steel	1.00–2.00% C	Very high hardness, limited ductility	Special tools and wear-resistant parts

8.3 Microstructure, Properties and Applications of Steels

Steel type	Typical microstructure	Important properties	Applications
Low-carbon steel	Ferrite + small pearlite	Soft, ductile, weldable	Sheets, bolts, nuts
Medium-carbon steel	Ferrite + pearlite	Strong, tough after heat treatment	Shafts, axles, gears
High-carbon steel	Pearlite + cementite	Hard, wear resistant	Springs, tools, dies
Alloy steel	Depends on alloying and heat treatment	Improved strength, hardenability, corrosion resistance or heat resistance	Machine parts, tools, turbines

9 Classification of Cast Irons

9.1 Definition of Cast Iron

Cast iron is an iron-carbon alloy containing more than 2.11 wt.% carbon. Commercial cast irons usually contain about 2.5–4.0 wt.% carbon along with silicon and other elements. Cast irons have good castability and are widely used for complex shapes.

TYPES OF CAST IRON			
Gray Cast Iron	White Cast Iron	Ductile (Nodular) Cast Iron	Malleable Cast Iron
CARBON FORM  Graphite flakes	CARBON FORM  Cementite (iron carbide)	Spheroidal (nodular) graphite Added element: Magnesium or cerium 	CARBON FORM Temper carbon (nodules after heat treatment) Made from: White cast iron via annealing
PROPERTIES Good machinability, vibration damping, poor tensile strength 	PROPERTIES Hard, brittle, wear-resistant, poor machinability Wear-resistant surfaces, mill liners, crushing rollers 	High strength, toughness, ductility USES Automotive components, gears, crankshafts 	PROPERTIES Strong, ductile, shock-resistant 

Figure 14: Classification of cast irons.

9.2 Gray Cast Iron

Gray cast iron contains carbon mainly in the form of **flake graphite**. The fracture surface appears gray due to the presence of graphite flakes.

Microstructure: Graphite flakes in ferrite, pearlite or ferrite-pearlite matrix.

Properties:

- Good machinability.
- High damping capacity.
- Good compressive strength.
- Low tensile strength and low ductility due to graphite flakes.

Applications: Machine tool beds, engine blocks, cylinder heads, pipes and housings.

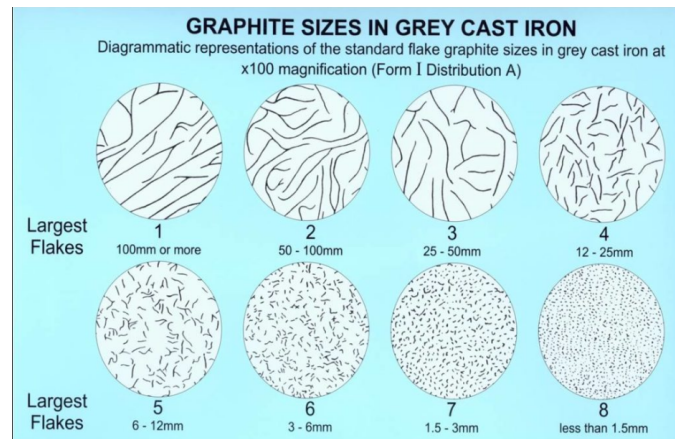


Figure 15: Gray cast iron showing flake graphite.

9.3 White Cast Iron

White cast iron contains carbon mainly as **cementite** instead of graphite. It has a white crystalline fracture surface.

Microstructure: Cementite and pearlite; often contains ledeburite in high-carbon regions.

Properties:

- Very hard and wear resistant.
- Very brittle.
- Difficult to machine.

Applications: Wear plates, liners, grinding balls and rolls.

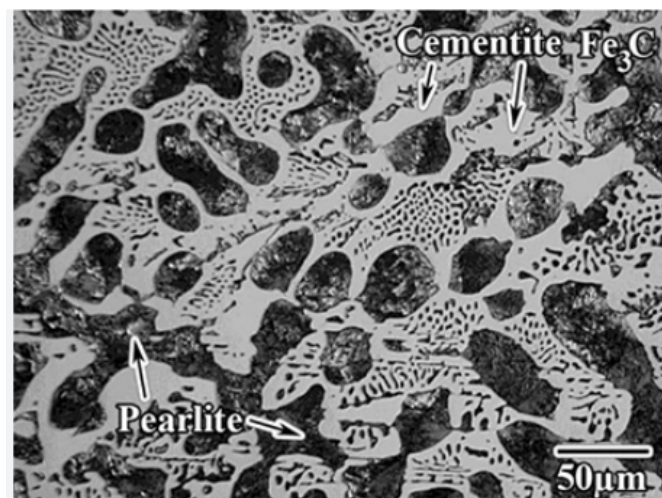


Figure 16: White cast iron showing cementite-rich microstructure.

9.4 Chilled Cast Iron

Chilled cast iron is produced when the surface of molten cast iron is cooled rapidly using a metallic chill. The rapidly cooled surface becomes hard white cast iron, while the inner region may remain gray cast iron.

Properties: Hard wear-resistant surface and tougher core.

Applications: Railway wheels, rolls and crushing equipment.

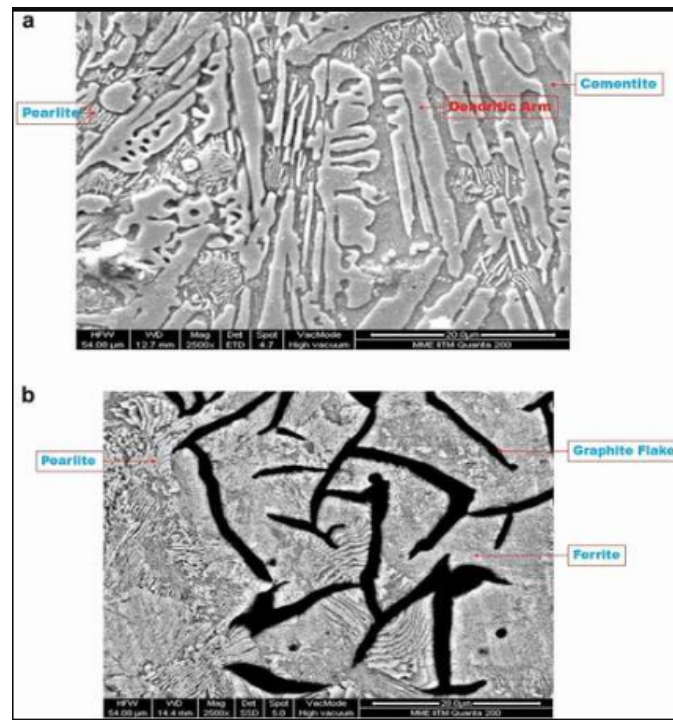


Figure 17: Chilled cast iron showing hard outer chilled layer.

9.5 Ductile Cast Iron

Ductile cast iron, also called nodular cast iron or spheroidal graphite iron, contains graphite in the form of nodules. Magnesium or cerium is added to promote nodular graphite formation.

Microstructure: Graphite nodules in ferritic, pearlitic or ferrite-pearlite matrix.

Properties:

- Higher ductility than gray cast iron.
- Good strength and toughness.
- Better impact resistance.
- Good machinability.

Applications: Pipes, crankshafts, gears, automotive suspension parts and pressure components.

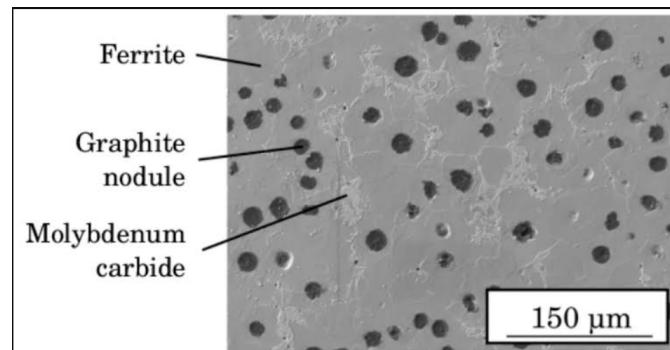


Figure 18: Ductile cast iron showing nodular graphite.

9.6 Malleable Cast Iron

Malleable cast iron is produced by heat treating white cast iron for a long time. During heat treatment, cementite decomposes and forms temper carbon nodules.

Microstructure: Temper carbon nodules in ferritic or pearlitic matrix.

Properties:

- Better ductility than white cast iron.
- Good shock resistance.
- Good machinability.
- Moderate strength.

Applications: Pipe fittings, brackets, agricultural parts and small machine components.

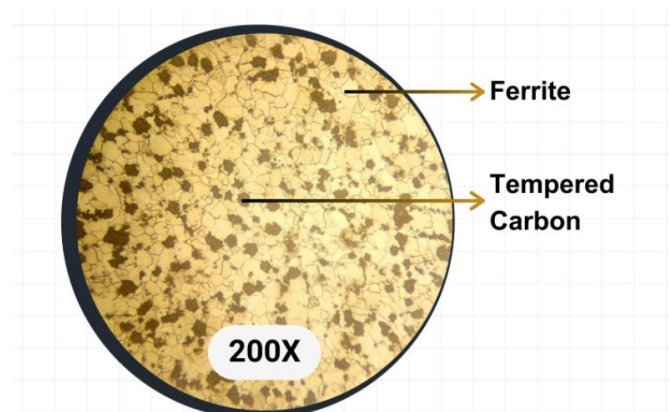


Figure 19: Malleable cast iron showing temper carbon nodules.

9.7 Cast Irons - Comparison

Type	Carbon form	Main properties	Applications
Gray cast iron	Flake graphite	Good machinability, high damping, brittle in tension	Machine beds, engine blocks
White cast iron	Cementite	Very hard, wear resistant, brittle	Liners, rolls, grinding balls
Chilled cast iron	Surface cementite + inner graphite	Hard surface and tougher core	Railway wheels, rolls
Ductile cast iron	Nodular graphite	Strong, tough, ductile	Pipes, gears, crankshafts
Malleable cast iron	Temper carbon	Ductile, machinable, shock resistant	Pipe fittings, brackets

10 Important Exam-Oriented Points

10.1 Very Short Answer Points

1. Alloy is a metallic material formed by combining two or more elements.
2. Solid solution is a single solid phase containing solute atoms dissolved in a solvent lattice.
3. Substitutional solid solution forms when solute atoms replace solvent atoms.
4. Interstitial solid solution forms when small solute atoms occupy spaces between solvent atoms.
5. Phase diagram shows phases present at different temperature and composition.
6. Isomorphous system has complete solubility in liquid and solid states.
7. Eutectic reaction: $L \rightarrow \alpha + \beta$.
8. Eutectoid reaction in steel: $\gamma \rightarrow \alpha + \text{Fe}_3\text{C}$.
9. Cementite contains 6.67 wt.% carbon.
10. Steel contains carbon up to about 2.11 wt.%.
11. Cast iron contains carbon above about 2.11 wt.%.
12. Gray cast iron contains flake graphite.
13. Ductile cast iron contains nodular graphite.

10.2 Important Two-Mark Questions

1. Define alloy and give two examples.
2. Define phase and component.
3. What is a solid solution?
4. Differentiate substitutional and interstitial solid solution.
5. What is an isomorphous phase diagram?
6. Write the eutectic and eutectoid reactions.
7. Mention the important phases in the Iron–Carbon diagram.
8. What is pearlite?
9. Define steel and cast iron based on carbon content.
10. Mention two applications of gray cast iron.

10.3 Important Five-Mark Questions

1. Explain substitutional and interstitial solid solutions with neat sketches.
2. Explain important terms used in phase diagrams.
3. Explain the Cu–Ni isomorphous phase diagram.
4. Explain eutectic, eutectoid, peritectic and peritectoid reactions.
5. Classify steels based on carbon content and mention applications.
6. Explain the types of cast irons with microstructure and applications.

10.4 Important Long-Answer Questions

1. Explain the Iron–Iron carbide equilibrium diagram with important phases, temperatures and reactions.
2. Classify steels and cast irons. Explain their microstructure, properties and applications.
3. Explain phase diagrams and invariant reactions with neat sketches.

11 Unit Summary

- Alloys are formed to improve the properties of pure metals.
- Solid solutions are classified as substitutional and interstitial.
- Phase diagrams help to predict phases, microstructures and transformation temperatures.
- The Cu–Ni system is a common example of an isomorphous system.
- Eutectic, eutectoid, peritectic and peritectoid reactions are important invariant reactions.
- The Iron–Iron carbide diagram is essential for understanding steels and cast irons.
- Steels are classified based on carbon content and alloying elements.
- Cast irons are classified based on the form of carbon and microstructure.

Glossary of Important Terminologies

The following short definitions are useful for quick revision and two-mark examination answers.

Term	Short definition / exam point
Alloy	A metallic material formed by combining two or more elements, of which at least one is a metal.
Base metal	The major metallic element present in an alloy.
Alloying element	An element added to a base metal to improve properties such as strength, hardness, corrosion resistance or wear resistance.
Constitution of alloy	The phases, microstructures and chemical distribution present in an alloy at a given temperature and composition.
Phase	A physically and chemically homogeneous portion of a material separated from other portions by a boundary.
Component	An independent chemical species used to describe the composition of a system.
System	A selected material or combination of materials considered for metallurgical study.
Solvent	The major element in a solid solution.
Solute	The minor element dissolved in the solvent lattice.
Solid solution	A single solid phase in which solute atoms are dissolved in a solvent crystal lattice.
Substitutional solid solution	A solid solution in which solute atoms replace solvent atoms in the crystal lattice.
Interstitial solid solution	A solid solution in which small solute atoms occupy interstitial spaces between solvent atoms.
Hume-Rothery rules	Conditions such as atomic size, crystal structure, electronegativity and valency that favour solid solubility.
Phase diagram	A graphical representation showing phases present under different temperature and composition conditions.
Equilibrium	A stable condition in which no further phase change occurs with time under fixed conditions.
Liquidus line	The boundary above which an alloy is completely liquid.
Solidus line	The boundary below which an alloy is completely solid.
Solvus line	A boundary showing the limit of solid solubility in a phase.
Tie line	A horizontal line drawn across a two-phase region to determine the composition of phases at a given temperature.

Lever rule	A rule used to calculate the relative amount of phases present in a two-phase region.
Isomorphous system	A binary alloy system having complete solubility in both liquid and solid states.
Invariant reaction	A phase reaction that occurs at a fixed temperature and composition in a binary phase diagram.
Eutectic reaction	A reaction in which one liquid phase transforms into two solid phases: $L \rightarrow \alpha + \beta$.
Eutectoid reaction	A reaction in which one solid phase transforms into two different solid phases: $\gamma \rightarrow \alpha + \text{Fe}_3\text{C}$ in steel.
Peritectic reaction	A reaction in which a liquid and a solid phase combine to form a new solid phase: $L + \alpha \rightarrow \beta$.
Peritectoid reaction	A reaction in which two solid phases combine to form a new solid phase: $\alpha + \beta \rightarrow \gamma$.
Iron–Iron carbide diagram	An equilibrium phase diagram used to study steels and cast irons in the Fe–Fe ₃ C system.
Ferrite	Soft and ductile body-centred cubic iron phase with very low carbon solubility.
Austenite	Face-centred cubic iron phase with higher carbon solubility, stable at elevated temperatures.
Cementite	Iron carbide, Fe ₃ C, containing 6.67 wt.% carbon; hard and brittle.
Pearlite	A lamellar mixture of ferrite and cementite formed by eutectoid transformation.
Ledeburite	A eutectic mixture formed in the Fe–Fe ₃ C system at 4.3 wt.% carbon and 1147°C.
Eutectoid steel	Steel containing approximately 0.8 wt.% carbon.
Hypo-eutectoid steel	Steel containing less than 0.8 wt.% carbon, generally having ferrite and pearlite.
Hyper-eutectoid steel	Steel containing more than 0.8 wt.% carbon, generally having pearlite and cementite.
Steel	An iron-carbon alloy containing up to about 2.11 wt.% carbon.
Cast iron	An iron-carbon alloy containing more than 2.11 wt.% carbon, usually with good castability.
Low-carbon steel	Steel containing about 0.05–0.30 wt.% carbon; soft, ductile and weldable.
Medium-carbon steel	Steel containing about 0.30–0.60 wt.% carbon; stronger and suitable for shafts, gears and axles.

High-carbon steel	Steel containing about 0.60–1.00 wt.% carbon; hard and wear resistant.
Gray cast iron	Cast iron in which carbon is mainly present as flake graphite.
White cast iron	Cast iron in which carbon is mainly present as cementite, giving high hardness and brittleness.
Chilled cast iron	Cast iron with a rapidly cooled hard surface and relatively tougher core.
Ductile cast iron	Cast iron containing nodular graphite, giving better ductility and toughness.
Malleable cast iron	Cast iron produced by heat treating white cast iron to form temper carbon nodules.
Graphite	A soft form of carbon present in many cast irons.
Microstructure	The internal structure of a material observed under a microscope, consisting of phases and grains.
Grain	A small crystal region within a polycrystalline metal.
Grain boundary	The boundary separating two neighbouring grains having different crystal orientations.
Mechanical properties	Properties such as strength, hardness, ductility, toughness and wear resistance.
